

Week 1 Reading: Process Prediction

1. Define the map before collecting data

A surrogate predicts outputs from specified inputs over a declared domain. A label is a reference output paired with one input row.

Role	This prediction task
Inputs	T: 300–360 K; τ : 1–10 min; C_{Af} : 0.8–1.5 mol/L
Outputs	C_A in mol/L; conversion X , dimensionless
Fixed context	Ideal, constant-volume, isothermal CSTR; first-order $A \rightarrow B$; fixed rate parameters
Reference	Steady-state material-balance solution

Feed concentration is an input here. It is not fixed to 1.2 mol/L for all data. Reaction assumptions and units must stay the same across the rows.

Check: a number called “temperature” is insufficient unless its units and operating range are known.

2. Derive the steady-state map

The ideal CSTR is perfectly mixed, isothermal, and constant-volume, with first-order reaction $A \rightarrow B$. The balance is accumulation = inlet – outlet – reaction:

$$\frac{dC_A}{dt} = \frac{C_{Af} - C_A}{\tau} - k(T)C_A$$

At steady state, set the derivative to zero. Multiply by residence time, move the concentration terms to the same side, and factor out outlet concentration. Dividing by the positive factor gives:

$$C_A = \frac{C_{Af}}{1 + k(T)\tau}, \quad X = 1 - \frac{C_A}{C_{Af}} = \frac{k(T)\tau}{1 + k(T)\tau}$$

At 330 K, the rate constant is 0.2/min. At 5 min residence time, the product of rate and residence time is 1, so conversion is 0.5. A 1.2 mol/L feed gives a 0.6 mol/L outlet.

The conversion is independent of feed concentration in this particular first-order model. That property is not a general rule for all reactors.

3. Build independent train, validation, and test data

Use this teaching rate law, with T in K and k in 1/min:

$$k(T) = 0.2 \exp \left[6000 \left(\frac{1}{330} - \frac{1}{T} \right) \right]$$

Sample inputs independently and uniformly inside the declared box; compute each label with the steady-state equations. Seed 42 fixes a reproducible sample.

Split	Rows	Role
Train	400	Fit coefficients
Validation	120	Compare model choices
Test	160	Final held-out evaluation

The provided basis is fixed in advance. If comparing another basis or capacity, use validation and leave test untouched. A new simulator sample is not automatically a new experimental campaign.

4. Scale inputs and fit ten features

$$\tilde{x}_j = 2 \frac{x_j - l_j}{u_j - l_j} - 1$$

At [330 K, 5 min, 1.2 mol/L], the scaled input is approximately [0, -0.111111, 0.142857]. The scaling uses declared bounds, not statistics computed from test data.

The feature row contains 1 constant + 3 inputs + 3 squares + 3 pairwise products = **10 features**. For 400 train rows, Φ has shape (400, 10), Y (400, 2), and B (10, 2).

$$\min_B \|\Phi B - Y\|_F^2, \quad \hat{y} = \varphi(x) B$$

`np.linalg.lstsq` fits B . Keep B fixed when predicting new input rows. Fitting parameters is an optimization calculation; selecting plant operating conditions is a later, separate problem.

5. Read accuracy and consistency separately

$$\text{RMSE}_j = \sqrt{\frac{1}{N} \sum_{n=1}^N (\hat{y}_{nj} - y_{nj})^2}$$

For each output, subtract the reference from the prediction, square, average over the test rows, and take the square root. Concentration RMSE is in mol/L; conversion RMSE is dimensionless.

The seed-42 baseline gives concentration RMSE **0.032623 mol/L** and conversion RMSE **0.025608**. Inspect individual errors as well as the averages.

$$r = \hat{C}_A + C_{Af}\hat{X} - C_{Af}$$

For feed 1.2, prediction $\hat{C}_A = 0.65$ and $\hat{X} = 0.45$ gives $r = 0.65 + 1.2 \times 0.45 - 1.2 = -0.01$ mol/L. The predicted pair is not exactly consistent with the defined conversion relation. Physical-consistency methods are studied in Week 4.

6. Answer checks and readings

Try first: at 330 K, 10 min, and feed 1.2 mol/L, what are C_A and X ? How many regression coefficients are fitted for two outputs? Why can a low training error be misleading?

Answer check: $k\tau = 2$, so $C_A = 1.2/3 = \mathbf{0.4 \text{ mol/L}}$ and $X = \mathbf{2/3}$. B contains **20 coefficients**: ten features for each of two outputs. Training evaluates the same points used for fitting; independent errors can be larger.

Work through this course-specific companion and the [Week 1 lab](#). Optional background: Boyd and Vandenberghe, *Convex Optimization*, affine maps and least-squares problems in Chapters 2 and 4 ([official book PDF](#)).

Week 2 introduces MLPs; Week 3 adds dynamic prediction. Operating objectives and constraints return in Weeks 5–8.